Supplementary Material:
Every Fact, and how it was obtained

for *The Internal Space and Boundary Growth of the Spectre and Tsiokos's Chair44 Tilings*

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What this is

The main paper stratifies its statements: *Facts* are exact statements, each proved, verified by an exact finite computation over a stated range, or quoted; *Propositions* carry a modelling step. The numbered Facts of the paper (25 of them) are drawn from a ledger of finer entries. This document lists every ledger entry the paper draws on — all 71 — with the key under which the ledger (`spectrefacts.py`, `chair44facts.py`, `monotilefacts.py`) computes it, the claim, the first sentence of the method by which it was obtained, and a mark on the 24 entries that name a theorem of the Lean 4 file. The ledger is shared with other documents; their entries are not listed here.

Three things about the table.

- **It is generated, not written.** Every row is emitted at build time from `spectrefacts.py`, the same module that produces every number in the main paper. There is no second copy to drift.
- **The Lean column is measured, not asserted.** A checkmark means the Fact's `lean` field names a theorem that exists in `SpectreFacts.lean` — 67 theorems, kernel-accepted, no admitted proofs — and the paper's build fails if a named theorem is missing; six phantom attributions were found and removed when that check was added. A Lean theorem proves the Lean statement, which can be narrower than the paper's: the entries marked * are proved in Lean for the sequence that the Lean file *defines* by the perimeter recurrence. That this sequence is the supertile perimeter at every order is Theorem 1 of the paper, proved by an exact finite certificate (`boundary_certificate/`), not in Lean.
- **The method column is the first sentence.** A method that runs to a paragraph is truncated at its first full stop or semicolon. The full text is in the module, one lookup by key away.

One entry is an input rather than a result, and the paper says so: `charpoly` rests on the transcribed substitution matrix (the published rule). It is audited against a second implementation of the same substitution, and an audit is not a proof. The perimeter recurrence, `boundary_recurrence`, was once in the same position, counted from placed tiles for $k \le 6$; it is now proved for every k by the finite certificate of Theorem 1 (`boundary_certificate/`), and its row says so.

The table

Generated from `spectrefacts.py`. The last column marks the Facts that are also proved in Lean, see https://github.com/brentharts/spectre/wiki/Spectre-Lean; the others are computations.

Key	Claim	How	Lean
`charpoly`	$\chi_M(x) = x^5(x-1)(x+1)(x^2-8x+1)$	exact characteristic polynomial over Z, factored	
`perron`	$\lambda^2 = 4 + \sqrt{15}$	largest root of the quadratic factor	
`pell`	$4^2 - 15 \cdot 1^2 = 1$	the Pell equation, so lambda^2 is the fundamental unit of Z[sqrt15]	✓
`lam_closed`	$\lambda = \frac{1}{2}(\sqrt{6} + \sqrt{10})$	square both sides: (sqrt6+sqrt10)^2/4 = (16+4 sqrt15)/4	
`minpoly`	$x^4 - 8x^2 + 1$	minimal polynomial of lambda over Q	
`channel_hex`	$(\lambda - \lambda^{-1})^2 = 6$	lambda^2 + lambda^-2 - 2, in Z[sqrt15], no root evaluated	✓

Key	Claim	How	Lean
channel_pent	$(\lambda+\lambda^{-1})^2=10$	lambda^2 + lambda^-2 + 2, in Z[sqrt15], no root evaluated	✓
contraction	$\lambda^{-4}=31-8\sqrt{15}$	the square of the conjugate unit	✓
jordan	$\operatorname{rank}M=5$, $\operatorname{rank}M^n=4$ $(n\geq 2)$	exact elimination over Q	
q_plus_eigen	$Q_+M=Q_+$	left eigenvector for the marginal eigenvalue +1	✓
q_minus_eigen	$Q_-M=-Q_-$	left eigenvector for the marginal eigenvalue -1	✓
q_values	$Q_+\equiv -1,\quad Q_-=+1,-1,+1,-1$	evaluated on the matrix census at depths one to four	✓
censuses	$T=1,8,63,496$; $T_{\text{geom}}=1,9,71,559$	the same patches counted with and without the Mystic	
mystic_rule	$T_{\text{geom}}(n)=T(n)+T(n-1)$	row Gamma of M is all ones, so N_Gamma(n) is the whole census at n-1 and the Mystic is a second copy of it	✓
ledger	$N_\Theta(n)=N_\Gamma(n-1)$	row Theta of M is the indicator of Gamma, so the count is the previous generation of Gamma verbatim	
ledger_lambda	$N_\Lambda(n)=N_\Sigma(n-1)$	row Lambda of M is the indicator of Sigma	
hat_charpoly	$\chi_H(x)=(x-1)(x+1)(x^2-7x+1)$	characteristic polynomial of the four-metatile Hat substitution	
hat_perron	$\kappa=\frac{1}{2}(7+3\sqrt{5})=\varphi^4$	largest root of the Hat quadratic, and the fourth power of the golden ratio	
phase_fields	$\mathbb{Q}(\sqrt{15})$ against $\mathbb{Q}(\sqrt{5})$	the two characteristic polynomials differ in a single coefficient, seven against eight, and that...	
frequencies	$v\in\mathbb{Z}[\sqrt{15}]^9$, $\sum v=1$	nullspace of M - lambda^2 I, normalised	✓
hat_frequencies	$f_H=\frac{1}{3}$; $f_T, f_P, f_F\notin\mathbb{Q}$	Perron eigenvector of the four-metatile Hat substitution, over Q(sqrt5)	
aperiodicity_witness	9/9 vs 3/4 irrational	a periodic tiling has rational frequencies, so an irrational one refutes periodicity	✓
trace_accounting	tr =unit+conjugate: 8, 7	the marginal pair +1 and -1 cancels and the zero modes contribute nothing, so the trace is the unit plus...	✓
twist_two_levels	$\chi_B(x)=(x-1)^2(x^2-8x+1)$, $B\,v=\lambda^2 v$	the supertile quadrilateral moves by an exact map, transcribed from the construction	
spectre_twist	$\tan\theta_S=\frac{1}{13}(5\sqrt{3}-6)$, $\cos 2\theta_S=\frac{1}{10}(4+3\sqrt{3})$	in the limit of high level, one level relates the quadrilaterals by ratio lambda and rotation minus...	
twist_field	$\cos 2\theta_S\in\mathbb{Q}(\sqrt{3})$, $\notin\mathbb{Q}(\sqrt{15})$	the angle lies in the field of the hexagonal tile coordinates	
boundary_recurrence	$P(k)=5P(k-1)-3P(k-2)-P(k-3)$	proved for every k by the finite certificate boundary_certificate/check_spectre_boundary_proof.py (93...	✓*
boundary_chi	$\chi_\partial(x)=(x-1)(x^2-4x-1)$	characteristic polynomial of the companion matrix of the recurrence	
boundary_growth	$\nu=2+\sqrt{5}=\varphi^3$	Perron root of the boundary operator	
boundary_field	$\nu\in\mathbb{Q}(\sqrt{5})$, $\lambda^2\in\mathbb{Q}(\sqrt{15})$	area and perimeter of the same tile inflate in different quadratic fields	
boundary_dimension	$\dim_B\partial K=\log\nu/\log\lambda$	box dimension of the boundary of the limiting supertile, proved by the certificate...	
balanced_exponent	$s^*=1-\log\nu/\log\lambda^2$	the thickness exponent making every substitution level carry comparable volume (a modelling choice)	
compositum	$\mu=\lambda^2/\nu$ has degree 4	the balanced ratio lies in Q(sqrt3, sqrt5) and in neither quadratic field alone	
golden_absent	$(x^2-4x-1)\nmid\chi_M$, χ_H	the boundary quadratic divides neither substitution matrix	
lucas_deflation	$\varphi^n:\ x^2-L_nx+(-1)^n$	every power of the golden ratio has a Lucas quadratic for its minimal polynomial, so agreement of two such...	
spectre_not_golden	$\lambda^2\neq\varphi^n$	the trace of the Spectre unit is eight, which is not a Lucas number, so the area inflation is not golden...	
boundary_invariant	$P(k+2)-4P(k+1)-P(k)=-16$	the quadratic factor of chi applied to the sequence annihilates the growing part and leaves the constant...	✓*
boundary_growth_bound	$P(k+3)\geq 4P(k+2)+30$	for the sequence defined by the recurrence (proved in Lean) the perimeter more than quadruples at every...	✓*

Key	Claim	How	Lean
`lam_below_four`	$\lambda^2 < 16$	16 - lambda^2 = 12 - sqrt15, positive since 12^2 ¿ 15	✓
`boundary_exact`	$P(k)$ counted exactly in $\mathbb{Q}(\sqrt{3})$	the same sequence recounted with edges identified by equality of exact coordinates in Z[zeta_12] rather...	
`chair_mesh`	$(V, E, T) = (2138, 6408, 4272)$, $\chi = 2$	the 24 panels of the seven-cube chair, each cut by its 9x9 grid into 73 flat rectangles and 8 four-sided...	✓
`chair_volume`	vol $= 7 + \frac{4}{3}w^2 h \sum_s \gamma_s = 7$ for all w, h	the volume of the symbolic solid, for every base width w and height unit h at once: the signed heights sum...	✓
`chair_labels`	192 sites, $\gamma_s \in \{\pm 1, \ldots, \pm 12\}$, $\sum \gamma_s = 0$	height labels forced by exact complementarity on every shared face of the depth-two patch of the recorded...	✓
`chair_tilings`	$3^8 = 6561$ rotation tilings of the doubled chair by 8 chairs	exact cover of the 56 cells of the doubled chair by eight rotated copies, by a bitmask search	
`chair_screen`	6555 tilings leave 4 free classes; 6 leave 12, and 3 of those use all 24 orientations	every one of the 6561 tilings is glued to depth two	
`chair_atlas`	2388 raw contacts→ 44 legal→ 30 realized	every face contact of two chairs (48 frames, translations in a box), the legal ones kept by exact site...	✓
`chair_equations`	372 site equations, rank = 180, dim ker = 12	the distinct complement equations c_A + c_B = 0 of the 30 realized contacts, ranked exactly over Q	✓
`chair_chiral`	Sym(decorated) = {1}, while |Sym(chair)| = 6 = 3 + 3	the undecorated seven-cube chair has six symmetries, three of them mirrors, so it is achiral	
`chair_forced_chirality`	no mirror contact is legal: all 44 legal contacts have det = +1	of the 2388 raw face contacts 1194 place a mirror image against the root	
`chair_orientation`	$\chi(x) = x^3(x-8)(x-2)^{20}$ on the 24 rotation species	the orientation transfer matrix of the substitution on the 24 rotations	
`chair_surface`	surface= $24 \cdot 4^N$, shared= $12(8^N - 4^N)$, $B_c = 12 \cdot 4^N(2^c - 1)$	counted on the placed patch for N ¡= 4 with the face-directions exactly flat (4^(N+1) per +-x,+-y,+-z)	✓
`chair_exponents`	$D_b = 2$, $s^* = 1/3$, $\mu = 2$ ($\lambda = 2$, $\nu = 4$, $d = 3$)	the unified exponents D_b = log nu / log lambda, s* = 1 - D_b/d, mu = lambda^d / nu with the Chair44...	
`chair_twist`	supertile= 2·chair, twist= 0, $\frac{1}{8}\sum_i H_i = \frac{1}{4} I$	the 8 child chairs are laid on the doubled lattice and their cells compared with the doubled chair: equal,...	
`chair_nets`	surface= 24 squares, $(V, E, F) = (26, 48, 24)$, 261,434,325,600 spanning trees	the exposed unit faces of the seven-cube chair, their edge-adjacency graph (every edge shared by exactly...	
`chair_slices`	$\sum_j d_j = 2^n(1, 1, 1)$, layer interface= $4^{n+1} - 4 \cdot 2^n$	the 8^n chairs of the depth-n patch, n¡=4, each reduced to the centre g(1,1,1)+t of its bounding cube and...	
`orientation_sums`	$S_n^{\text{chair}} = 2^n$ exactly; $\lvert S_n^{\text{Spectre}}\rvert/\lambda^n \to 1.4489$	the sum of the orientation vectors over the depth-n patch: for the chair the z component of the notch...	
`power_slopes`	median $I(k)/N$ falls as $N^{-0.34}$ (Spectre) and $N^{-0.72}$ (chair)	the median over 300 fixed random wave vectors (seed 1) in the box [0, 2 pi / l]^d, with l the tile size...	
`chair_coincidence`	modular coincidence at depth 3, $c(M) = 28 \cdot 2^M - 48$ ($M \geq 3$)	the cell substitution on 24 x 7 = 168 labelled unit cells, rebuilt from the child poses: column sum 8,...	✓
`chair_legal_blocks`	1410 legal vertex blocks, closed under σ; 27 Φ-fixed seeds, stabilisers of order 1 and 8	the 2x2x2 windows of sigma^n(j) over the 168 labels, n = 1..6, from the cell substitution of...	
`chair_hull_theorem`	every atlas tiling lies in the hull; $\mathrm{Sym}(T) \in \{1, C_2^{\text{diag}}, C_4, D_4\}$, $\lvert\mathrm{Sym}(T)\rvert \leq 8$	Romanov (this paper): one level of the hierarchy plus a finite computation on the 8 cells around a lattice...	
`chair_lines`	box means of the height and notch observables are 0; 6 chair stacks give diagonal lines with means $\mp 36, \pm 1$	exact, from the cell substitution of chair44_twoadic	
`boundary_fibonacci`	$P(k) = 2F_{3k+5} + 4$	for the sequence defined by the perimeter recurrence (seeds 14, 46, 182, characteristic polynomial...	✓*
`boundary_directions`	even= $F_{3k+4} + 1$, odd= $F_{3k+3} + 1$ ($k \leq 6$)	boundary edges of an order-k supertile, sorted by direction mod 180 degrees (the 14 edges point in...	✓*
`boundary_z3_bounded`	$\max\lvert\hat{h}(j)\rvert$ over the Z/3 and faithful modes$\leq \sqrt{13}$ ($k \leq 6$)	the DFT of the folded direction histogram over Z/6 = Z/2 x Z/3 (the split spectre_torsion uses for the six...	
`centre_diffraction`	$X_n = 2X_{n-1} \cup$ (odd box\corner octant); $\sum_k I(k) = \frac{1}{7}$ =density	the centre of every chair of the depth-n patch obeys the nested recursion above (checked against the...	

Key	Claim	How	Lean
`orientation_recursion`	$S_{n+1} = C\,\overline{S_n}$, $\lvert S_n\rvert \sim c\,\lambda^n$, $c^2 = \frac{15+5\sqrt{3}+3\sqrt{5}+3\sqrt{15}}{20}$	the sum of exp(i angle) over the tiles of the depth-n supertile of each label obeys a linear recursion...	
`orientation_channels`	charpoly $= x^3(x-8)(x-2)^{20}$ (chair); $h_m = 2, 1, 0$ (Spectre)	the 24 orientation states of the Chair44 are split by the irreducible representations of the rotation...	
`contact_matrices`	Spectre: 14×14, rank 14, bipartite; Chair44: 24×24, rank 24, not bipartite	the number of times native edge a of one tile meets native edge b of a neighbour, counted over the level-4...	
`lift_integer`	every vertex has integer coordinates $(p, q, r, s) \in \mathbb{Z}^4 = \mathbb{Z}[\zeta_{12}]$; rank $= 4$	every edge of the patch is a unit vector at a multiple of 30 degrees: even multiples are p+qw with w=exp(i...	
`inflation_matrix`	charpoly $= (x-1)^4(x+1)^4(x^4-8x^2+1)^2$, $\lambda^2 = 4+\sqrt{15}$	the supertile recursion of spectre_patch moves four key points by rotations through multiples of 60...	
`internal_plane`	bounded plane over $\mathbb{Q}(\sqrt{15})$; window box count≈ 1.2 (exact seam dimension 1.111, by the window theorem)	a linear functional is bounded on the supertile corners exactly when it kills the expanding part...	